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Title: TRANS IQ: Iot-Embedded Based Truck Monitoring System for Driver Fatigue Detection, Temperature Analysis and GPS Tracking.

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Abstract: Fatigue, overheating and a lack of monitoring in truck drivers leads to truck accidents and cargo losses around the world. Conventional vehicle systems are not equipped with in real-time physiological and environmental monitoring. In this paper, we have developed TRANS IQ which is an IoT-based embedded system for driver's fatigue detection, real-time monitoring of the cabin temperature and live vehicle location tracking. The system has a wink detection sensor, temperature sensor and GPS module connected to an ESP32 microcontroller. Based on hybrid embedded-IoT architecture, the system analyses in real-time and alerts locally with a buzzer and LCD while sends data to the cloud-dashboard for fleet level monitoring. Experimental verification on a prototype truck model shows the effectiveness of the system in enhancing safety, response time and reliability of monitoring. The results show 96 % detection accuracy of drowsiness events, and a latency below 2 s for critical alerts. The SmartHalo design significantly lowers the cost, scalability and deployability implementation of safety in transportation with smart sensing and IoT integration.

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